Tutorial 1: Stationary Time Series

Empirical Exercises

1. The excel file USdata.xlsx contains the monthly time series of the US industrial production (IP), unemployment rate (UNR), money supply (M1) and month-on-month percentage change in the share prices (SP) over the period from Jan 1960 to Sep 2025. The dataset is collected from the Federal Reserve Economic Data (FRED).

   (a) Plot IP, UNR, M1 and SP. Which of them are likely stationary?

   (b) Define ΔIP = log(IP$_t$) - log(IP$_{t-1}$), Δ(UNR)=UNR$_t$-UNR$_{t-1}$, and Δ(M1)=log(M1$_t$)-log(M1$_{t-1}$). Plot these three time series using sample before COVID-19 pandemic (Dec 2019). Now, which of them are likely stationary.

2. Romer and Romer 2010 examines the effect of tax changes on real economic activitiy in the US. In the data file RRdata.xlsx you will find the dataset used in this paper.
   - $PGDP$: price index of real GDP
   - $PCGDPl$: first difference of the logarithm of real GDP, $PCGDPl = log(PGDP_t) - log(PGDP_{t-1})$.
   - $EXOGENRRATIO$: a ratio of the exogenous tax changes to nominal GDP at time t.

   (a) What is the economic meaning of PCGDPl? Plot log(PGDP) and PCGDPl. Based on the plot, which ones you think are stationary?

   (b) EXOGENRRATIO is a new measure of the exogenous fiscal shock that is essentially not serially correlated. One simple way to check the autocorrelations of lag order up to k=12 is to look at the autocorrelogram. Plot the autocorrelogram for the fiscal shock. Do they exhibit significant autocorrelation? Plot the autocorrelogram for PCGDPl (k=12) and make a comparison with that of EXOGENRRATIO.